

LM02 Alternative Investment Performance and Returns

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1. Introduction

This learning module covers:

- Issues in performance appraisal of alternative investments
- Calculating fees and returns of alternative investments

2. Alternative Investment Performance

Alternative investments differ from traditional asset classes in the following ways:

- Longer time horizons
- Unique patterns of cash flows
- The use of leverage
- Illiquid positions
- More complex fee structures
- Different tax and accounting treatment
- Less normally distributed returns

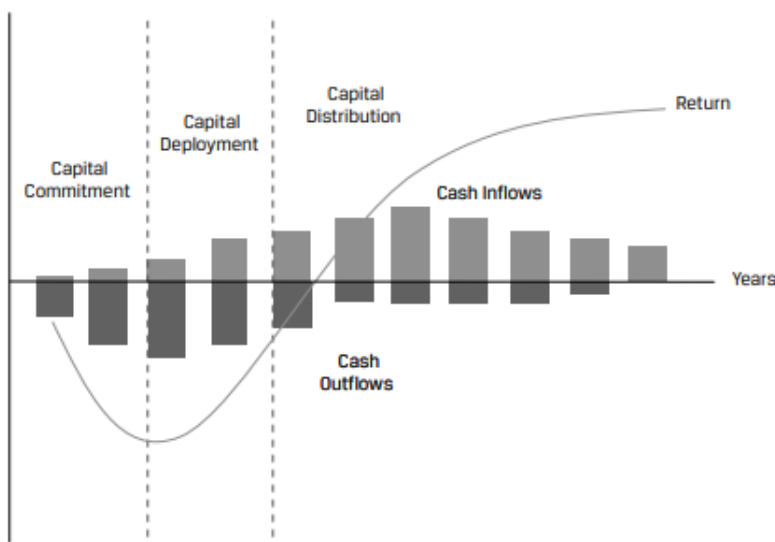
Due to these characteristics, it can be difficult to conduct performance appraisal on alternative investments.

When evaluating alternative investments, four factors should be considered:

- the life cycle phase of the investment
- the amount of borrowed funds used to maintain the market position
- the valuation of the assets
- the fee structure of the fund

Investment Life Cycle

Life cycle phases of various alternative investments generally fall into three distinct periods, as shown in Exhibit 1 from the curriculum.



Capital commitment: In this phase, managers identify and select appropriate investments with either an immediate or a delayed commitment of capital (known as a capital call). Returns are typically negative during this phase because fees and expenses are incurred immediately prior to capital deployment, and assets may generate little or no income during this period.

Capital deployment: In this phase, managers deploy funds to:

- construct or make property improvements (in case of real estate or infrastructure fund)
- incur expenses in the turnaround of a mature company (in case of private equity)
- initiate operations for a startup (in case of a venture capital)

Cash outflows typically exceed cash inflows. Management fees further reduce returns.

Capital distribution: If the property improvements/turnaround strategy/ or startup phase is successful, the underlying assets will appreciate in price and/or generate income exceeding costs. The fund can realize substantial capital gains from liquidating or exiting its investments.

Private Equity and Real Estate Performance Evaluation

Private equity and real estate investments often display a *J-curve effect* – initial decline followed by strong growth over the long term. This is because both private equity and real estate require significant initial cash outlays, and the investments take some time to turn profitable. Therefore, it is inappropriate to use short-term performance measures for private equity and real estate. Instead, the following measures are commonly used.

Performance measures for private equity:

The *IRR calculation* is frequently used to evaluate private equity investments. However, IRR has certain limitations as a performance measure for private equity:

- It doesn't take into account unrealized gains (since it only considers cash flows).
- It assumes that interim cash flows are reinvested at the same rate as the IRR of the investment (which may or may not actually be earned).
- And finally, it needs an assumption of the financing rate to use for outgoing cash flows to compare with the IRR to evaluate whether the return is more than what the required rate would be for the given amount of risk in the investment.

To overcome this complexity, the *multiple of invested capital (MOIC)*, or money multiple is frequently used. It simply measures the total value of all distributions and residual asset values relative to an initial total investment.

$$\text{MOIC} = (\text{Realized value of investment} + \text{Unrealized value of investment}) / (\text{Total amount of}$$

invested capital).

Although simple to calculate, a major drawback of MOIC is that it ignores the timing of the cash flows.

Example:

(This is Q#3 from the Question Set of the curriculum.)

A private equity closes a fund with a capital commitment of €750 million. It has a capital call of €500 million initially and another €250 million at the end of Year 1. The management fee is 2% per annum. At the end of Year 5, a total of €1.0 billion is distributed to its investors, and the fund is left with €500 million in asset value. Calculate the fund's MOIC.

Solution:

MOIC = (Realized value of investment + Unrealized value of investment)/Total amount of invested capital, where: invested capital equals total paid-in capital less management fees and fund expenses. MOIC is different from the IRR measure because it ignores the timing of cash flows.

Total paid-in capital = 500 + 250 = 750.

Total management fee for 5 years = 750 × 0.02 × 5 = 75.

Total invested capital = 750 – 75 = 675.

MOIC = (1,000 + 500)/675 ≈ 2.2×

Performance measures for real estate:

The *cap rate* is often used to evaluate real estate investments. It is calculated as the annual rent actually being earned divided by the price originally paid for the property.

Use of Borrowed Funds

Managers may use borrowed funds to increase investment returns. Leverage allows managers to take a market position that is larger than the capital committed.

Consider a cash investment V_c with a periodic rate of return r . If we assume an investor is able to borrow at a periodic rate of r_b to increase the size of its investment by borrowed funds of V_b , we can calculate a simple leveraged rate of return r_L for the period as follows:

$$r_L = \text{Leveraged portfolio return/Cash position} = [r \times (V_c + V_b) - (V_b \times r_b)]/V_c$$

This equation can be rearranged as:

$$r_L = r + V_b/V_c(r - r_b)$$

Hedge funds often use leverage to enhance returns. To lever their portfolio hedge funds use derivatives or borrow capital from prime brokers. Hedge funds must deposit cash or other collateral into a margin account with the prime broker, who then lends securities to the hedge fund. If the margin account falls below a certain threshold, a margin call is issued, and

the hedge fund is required to put up additional collateral. This can magnify a hedge fund's losses because it may have to liquidate the losing position to meet the margin call.

Example:

(This is based on Example 2 from the curriculum.)

A hedge fund with a \$100 million capital normally uses leverage to invest in a variety of equity-linked notes.

Scenario 1: Suppose the fund's underlying positions return 8%. If it could add leverage of USD50 million to the portfolio at a funding cost of 4%, what would have been the leveraged return?

The leveraged return can be calculated as follows:

$$V_c = 100; V_b = 50.$$

$$r_L = 0.08 + (50/100)(0.08 - 0.04) = 10\%$$

Scenario 2: Suppose the fund's underlying positions incur a loss of 2% instead of earning a gain. What would have been the leveraged return if the fund had borrowed USD50 million at 4%?

$$V_c = 100 \text{ and } V_b = 50$$

$$r_L = -0.02 + (50/100)(-0.02 - 0.04) = -5\%.$$

Valuation

Alternative investment funds may be valued on a daily, weekly, monthly, and/or quarterly basis. The value of a fund depends on the value of underlying positions.

The price used for valuation depends on whether market prices are available and if the underlying position is liquid. When market prices are available, the fund decides what price to use. Common practice is to quote at the average of the bid and ask prices. A conservative approach is to use bid prices for long and ask prices for short.

GAAP accounting rules categorize fund investments into three buckets:

- Level 1: An exchange-traded, publicly traded price is available and is used for valuation purposes.
- Level 2: When such price is not available, outside broker quotes are used.
- Level 3: When broker quotes are not available or are unreliable, as a final recourse, assets are valued using internal models.

Level 3 assets values require additional scrutiny from investors. The models used should be appropriate and consistent. The values obtained may not reflect true liquidation values. Also, the returns may be smoothed and the volatility understated.

Fees

The after-fee results of alternative investments can vary significantly based on 'which'

investor has invested 'when' in a particular fund.

For example:

- An investor may face significantly lower incentive fees if he invests more capital in a fund at an earlier phase or is willing to accept greater restrictions on redemptions.
- An investor who enters an alternative fund after it has experienced a sharp decline in value may incur performance fees if the fund rises, whereas an earlier investor who experienced the sharp decline in value from its peak may be exempt from such fees for the same period.

3. Alternative Investment Returns

Hedge funds have performance fees, and other customized, complex compensation arrangements that seek to align manager and investor incentives. These structures are designed to reward investors for early involvement, larger investments, and/or longer lockup periods.

Another factor that can magnify losses for hedge funds is redemption pressure. Redemptions usually occur when the hedge fund is performing poorly. Redemptions can force hedge fund managers to liquidate positions at disadvantageous prices.

To discourage redemptions:

- Hedge funds sometimes charge *redemption fees* (typically payable to the remaining investors) to offset the transaction costs for the remaining investors.
- Hedge funds use *notice periods* (investors need to inform the fund manager in advance before making a redemption) which provide the hedge fund manager an opportunity to liquidate positions in an orderly manner.
- Hedge funds use *lockup period* (time periods when investors cannot withdraw their capital) which provide the hedge fund manager sufficient time to implement his investment strategy.

Alternative Investment Returns

Hedge funds commonly use a "2 and 20" fee structure and fund of funds commonly use the "1 and 10" fee structure. However, many variations of the fee structure exist.

Analysts should be aware of any custom fee arrangements in place that will affect the calculation of fees and performance. These can include such arrangements such as:

- Fees based on liquidity terms and asset size: Hedge funds may provide a fee discount to investors who are willing to accept lower liquidity e.g., longer lockups. Similarly, hedge funds may provide a fee discount to larger investors. These terms are negotiated with individual investors via side letters, which are special amendments to the fund's LPA.
- Founder's share: To entice early participation in new hedge funds, managers often

offer incentives known as founder's class shares. These shares have a lower fee structure e.g., "1.5 and 10" instead of "2 and 20" and are typically applicable to a certain cutoff threshold e.g., the first \$100 million in assets.

- Either/or fees: A few large institutional investors have recently worked out a new fee model with some hedge fund managers. These managers agree either to charge a 1% management fee or to receive a 30% incentive fee above a mutually agreed-on hurdle rate, whichever is greater. The 1% management fee allows a fund to cover its expenses during down years and the 30% incentive fee incentivizes and rewards managers during up years.

Alternative Investment Return Calculations

Example: Incentive Fees Relative to Waterfall Types

A PE fund invests \$10 million in Portfolio company A and \$12 million in portfolio company B. Company A generates a \$6 million profit, but Company B generates a \$7 million loss. The time period for the gain and loss are the same. The manager's carried interest incentive fee is 20% of profits. Calculate the incentive fee under:

1. A European-style waterfall whole-of funds approach
2. An American-style waterfall deal-by-deal basis (assuming no clawback)

Solution to 1:

Overall, the fund lost money (+\$6 million - \$7 million = -\$1 million) so under a European-style whole-of-fund waterfall, the manager will not receive any incentive fee

Solution to 2:

Under an American-style waterfall, the GP could still earn $20\% \times \$6 \text{ million} = \1.2 million as incentive fees on the profitable Company A deal.

Example: Fee and return calculations

Consider a hedge fund with an initial investment of 200 million; the fee structure is 2 and 20 and is based on year-end valuation. In year 1, the return is 30%.

1. What is the total fee if management fee and incentive fee are calculated independently? What is the investor's effective return?
2. What is the total fee if the incentive fee is calculated after deducting the management fee? Investor's net return?
3. If there is a hurdle rate of 5% and fees are based on returns of in excess of 5%, what is the total fee? What is the investor's net return?
4. In the second year, the fund declines to 220 million. Assume that management fee and incentive fee are calculated independently as indicated in Part 1, but now a high-water mark is also used in fee calculations. What is the total fee? What is the investor's net return?

5. In the third year, the fund value increases to 256 million. What is the total fee and investor's net return?

Solution:

1. Initial investment grows to: $200 \times 1.3 = \$260$ million.

Profit = \$60 million.

Management fee: $0.02 \times 260 = \$5.2$ million.

Incentive fee which is 20% of profit = $20\% \times 60 = \$12$ million.

Total fee = \$5.2 million + \$12 million = \$17.2 million.

Investor's return = $\frac{260-17.2}{200} - 1 = 21.4\%$

2. Incentive fee after deducting management fee = $20\% \times (260 - 200 - 5.2) = 10.96$.

Total fee = $5.2 + 10.96 = \$16.16$ million.

Investor's return = $\frac{260-16.16}{200} - 1 = 21.92\%$.

As you can see the return is better than Part 1 because incentive fee paid is relatively less here.

3. There is a hurdle rate of 5%. So, $200 \times 0.05 = \$10$ million must be subtracted before incentive fees are paid.

Incentive fee = $0.2 \times (260 - 200 - 5.2 - 10) = 8.96$.

Total fee = $5.20 + 8.96 = \$14.16$ million.

Incentive fee is further reduced and the investor's return is enhanced.

Investor's return = $\frac{260-14.16}{200} - 1 = 22.92\%$.

4. Management fee = $0.02 \times 220 = 4.4$. To calculate the incentive fee, we need to determine whether the fund value has exceeded the high-water mark. The high-water mark was achieved at the end of Year 1. This value was $260 \text{ million} - 17.2 \text{ million} = 242.8 \text{ million}$. The incentive fee is 0 because the fund value is below the high-water mark. Hence the total fee = \$4.4 million.

Investor's return = $\frac{220-4.4}{242.8} - 1 = -11.2\%$

5. Management fee = $256 \times .02 = 5.12$. Since \$256 has exceeded high water mark of 242.8 million, an incentive fee would be paid. Incentive fee = $(256 - 242.8) \times 0.2 = 2.64$. Total fee = $5.12 + 2.64 = 7.76$ million.

Investor's net return = $\frac{256-7.76}{215.6} - 1 = 15.14\%$.

Example: Hedge fund versus fund of funds

An investor is contemplating investing £200 million in either the Hedge Fund (HF) or the Fund of Funds (FOF). FOF has a "1 and 10" fee structure and invests 10% of its assets under management in HF. HF has a standard "2 and 20" fee structure with no hurdle rate. Management fees are calculated on an annual basis on assets under management at the

beginning of the year. Management fees and incentive fees are calculated independently. HF has a 25% return for the year before management and incentive fees.

1. Calculate the return to the investor of investing directly in HF.
2. Calculate the return to the investor of investing in FOF. Assume that the other investments in the FOF portfolio generate the same return before management fees as HF and have the same fee structure as HF.

Solution to 1:

HF has a profit before fees on a £200 million investment of £50 million ($= 200 \text{ million} \times 25\%$). The management fee is £4 million ($= 200 \text{ million} \times 2\%$) and the incentive fee is £10 million ($= 50 \text{ million} \times 20\%$). The return to investor is 18% ($= (50 - 4 - 10) / 200$).

Solution to 2:

FOF earns an 18% return or £36 million profit after fees on £200 million invested with hedge funds. FOF charges the investor a management fee of £2 million ($= 200 \text{ million} \times 1\%$) and an incentive fee of £3.6 million ($= 36 \text{ million} \times 10\%$). The return to the investor is 15.2% ($= (36 - 2 - 3.6) / 200$).

Relative Alternative Investment Returns and Survivorship Bias

Hedge fund index returns can be overstated due to survivorship, and backfill biases.

- Survivorship bias occurs when an index is composed of only surviving funds over a period of time, which tends to overstate the index returns.
- Backfill bias occurs when a new fund enters a database and historical returns of that fund are added (i.e., “backfilled”). Usually, funds that performed well are added which tends to overstate the index returns.

Example: Clawbacks Due to Return Timing Differences

A PE fund makes two investments for \$5 million each in Company A and Company B. One year later Company A returns a \$8 million profit. But two years later Company B turns out to be a complete bust and is worth zero.

The GP's carried interest is 20% of aggregate profits and there is a clawback provision. How much carried interest will the GP receive in year 1 and year 2.

Solution:

In year 1, the GP will receive a carried interest of 20% of \$8 million = \$1.6 million. This amount would typically be held in an escrow account for the benefit of the GP but not actually paid.

In year 2, the GP loses \$5 million of the initial \$8 million gain, so the aggregate profit is only \$3 million. The carried interest payable is $20\% \times \$3 \text{ million} = \0.6 million . The GP has to return \$1 million of the previously accrued incentive fee to the LPs because of the clawback provision.

Summary

LO: Describe the performance appraisal of alternative investments.

It can be difficult to conduct performance appraisal on alternative investments because these investments have asymmetric risk–return profiles, limited portfolio transparency, illiquidity, product complexity, and complex fee structures.

The IRR and MOIC calculations are frequently used to evaluate private equity investments. The cap rate is frequently used to evaluate real estate investments.

Leverage, illiquidity and redemption pressure pose special challenges while evaluating hedge funds' performance.

LO: Calculate and interpret alternative investment returns both before and after fees.

Alternative investment managers usually charge a management fee based on AUM and an incentive fee based on performance. However, analysts should also be aware of any custom fee arrangements in place that will affect the calculation of fees and performance. These can include such arrangements such as: fees based on liquidity terms and asset size, founder's share, and either/or fees.

It is difficult to generalize performance appraisal for these investments because returns may vary depending on how and when a particular investor invested in a particular vehicle.

Hedge fund index returns can be overstated due to survivorship, and backfill biases.